Quality Management

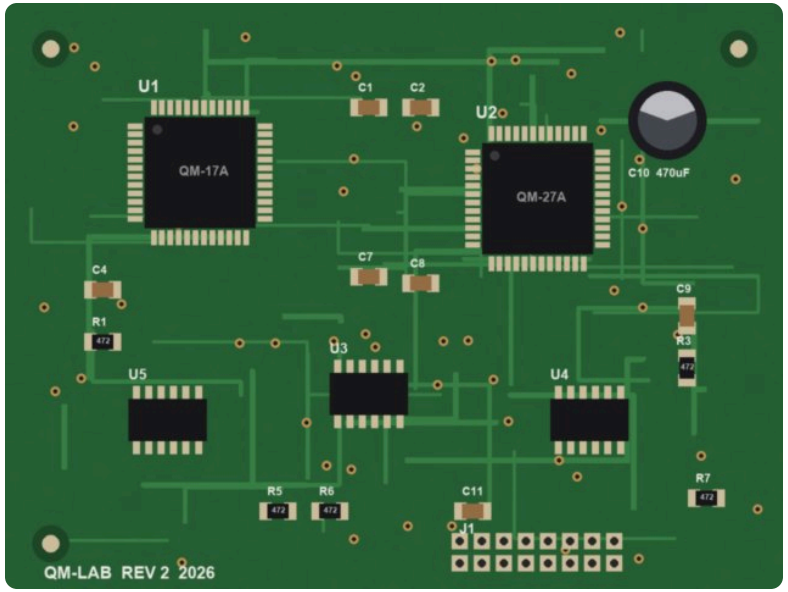
11 — Interest points: finding the same thing in two photos

Prof. Gianluca Rossi • **dr. Lorenzo Capponi**
BSc in Engineering Management • *Università degli Studi di Perugia*
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One board, two photos

Automatic inspection of circuit boards



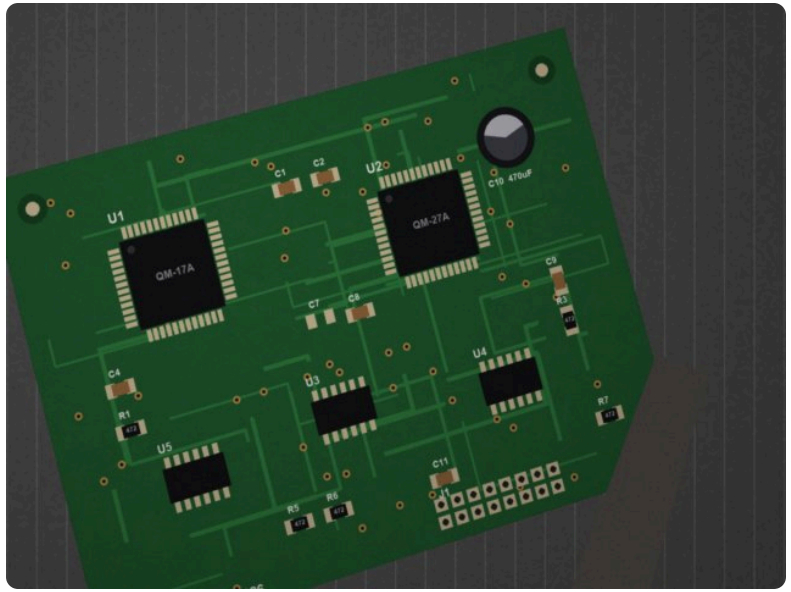


Photo 1 · the reference. A good board, photographed once.

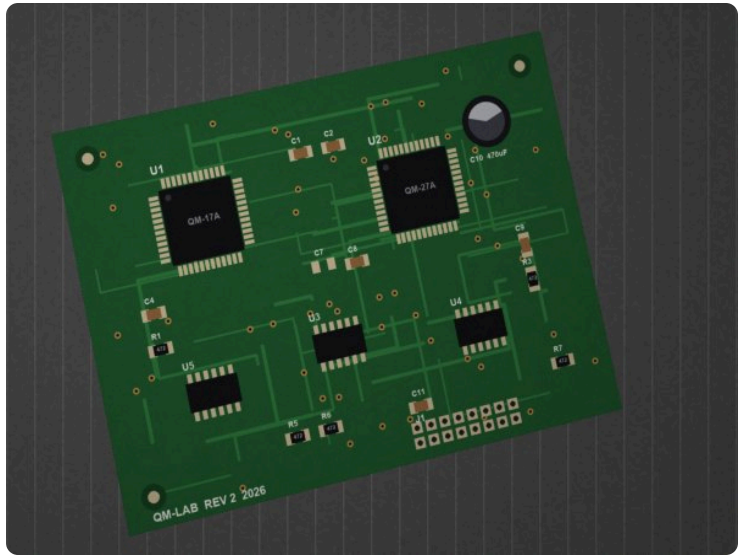
Photo 2 · the line. The next board on the conveyor: turned, farther away, darker, partly covered.

On the line, a camera photographs every board, and software compares it with the reference to find missing or wrong parts.

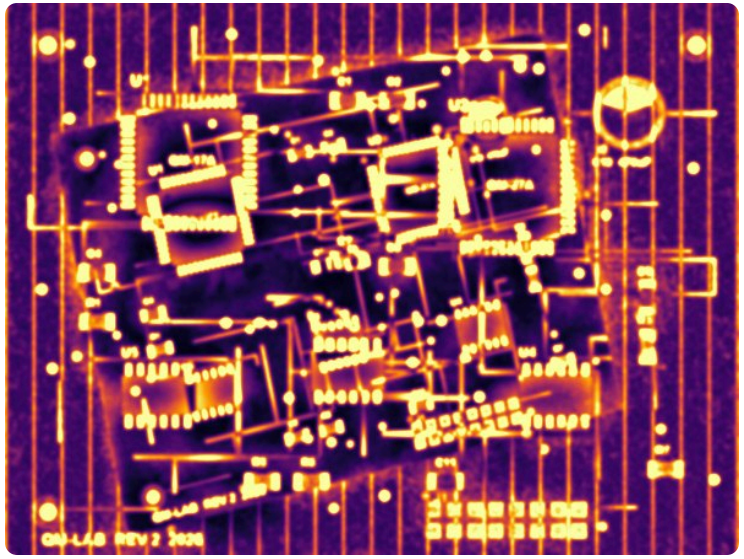
Everything in this lecture is computed on synthetic photos with OpenCV. The method (SIFT) is from D. G. Lowe, *International Journal of Computer Vision* 60(2), 2004; the teaching sequence follows S. K. Nayar, *First Principles of Computer Vision*.

Why not just compare the two photos?

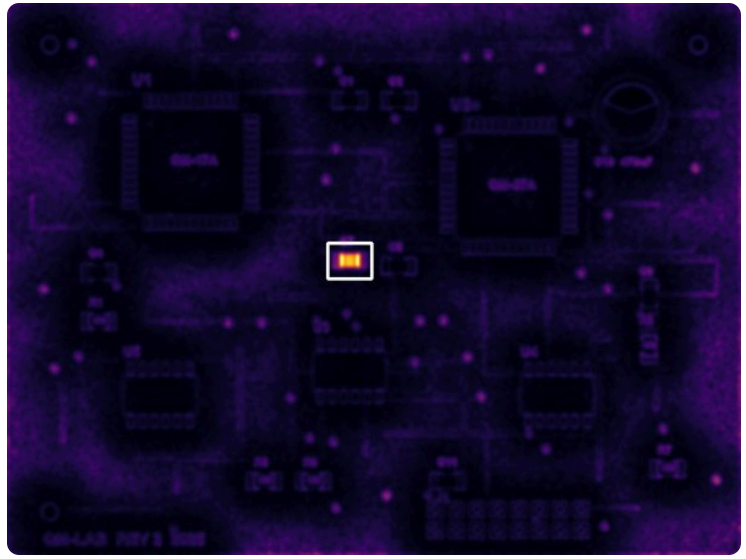
Subtract one photo from the other: what is left is what differs



A board on the line, as the camera sees it.



Subtracted directly: everything lights up. The boards do not line up, so every part looks different.



Subtracted after lining up: one spot is left — a missing capacitor.

Today's question

How does the software find **where the board is** in photo 2 — its position, angle and size?

The plan: four steps

The whole lecture in one picture — we come back to it at every step

1 · Find
the interest points

→

2 · Describe
each interest point

→

3 · Match
the two photos

→

4 · Use
the matches

where are they? how big? | *a direction and a fingerprint* | *pair them, drop the mistakes* | *align, inspect, measure*

Do not compare whole pictures. Find a few distinctive spots — **interest points**, the landmarks of an image — in both photos, recognise which is which, and work out how the board moved.

EVERYDAY · YOUR CAR IN A PHOTO

You find your car in a photo taken from the other side of the car park by a few details you know: the dent, the sticker, the roof box. Not by comparing every pixel.

INDUSTRY · THIS BOARD

The machine does the same with the corners of the chips, the pads and the capacitors — and then checks the board part by part.

PART 1

Why is this hard?

What the computer sees, and why the obvious idea fails

The computer sees only numbers

The same corner of chip U1, in the two photos

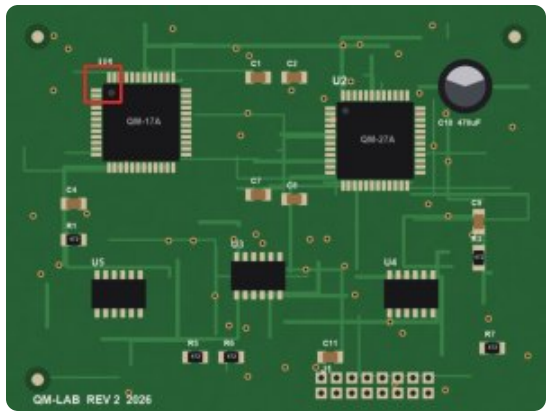


Photo 1 (red square: where)

73	73	73	72	72
73	71	64	61	61
73	64	46	37	37
72	61	37	26	26
72	61	37	26	24

Photo 1: grey levels, 0 = black, 255 = white

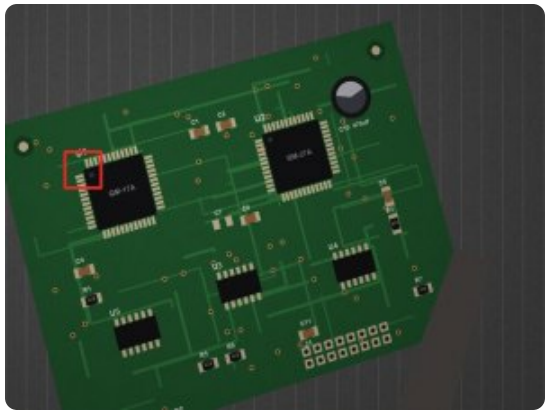


Photo 2 (red square: the same spot)

55	56	56	55	51
55	53	48	43	35
55	51	38	26	20
57	51	32	20	17
56	52	35	21	17

Photo 2: the same spot of the board

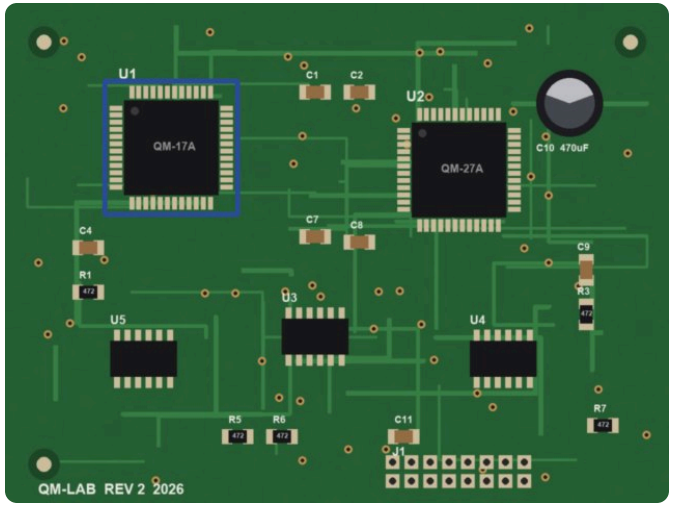
A photo is a grid of numbers (Lecture 8). The computer does not see “chip U1” — only these values. In photo 2 the board has turned, shrunk and darkened: **every single number has changed**.

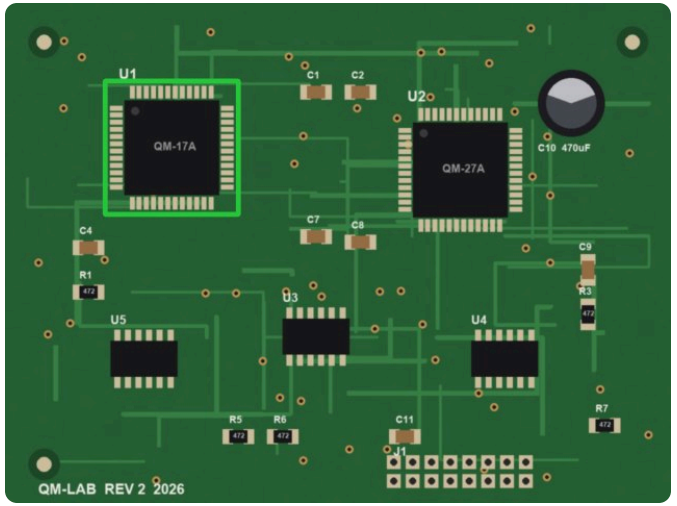
So the question is

What stays the same when all the numbers change?

Idea 1: search for a piece of photo 1

Template matching





1 · Cut chip U1 out of photo 1 (blue box) and keep it in the software: this small picture is the **template**.

2 · Slide the template over the new photo. At every position, score how similar the two are: 0 = nothing alike, 1 = identical.

3 · The best score is where the chip is. Board not turned: found (green), score **1.00**.

EVERYDAY

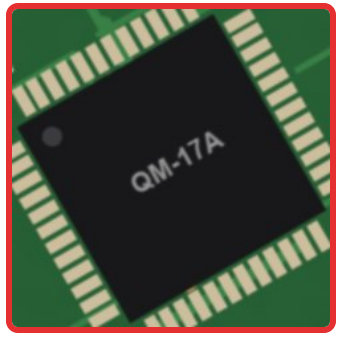
Like finding a word on a page by sliding a cut-out of that word over the text.

… and where it breaks

Turn the board, and the piece of photo 1 no longer looks the same

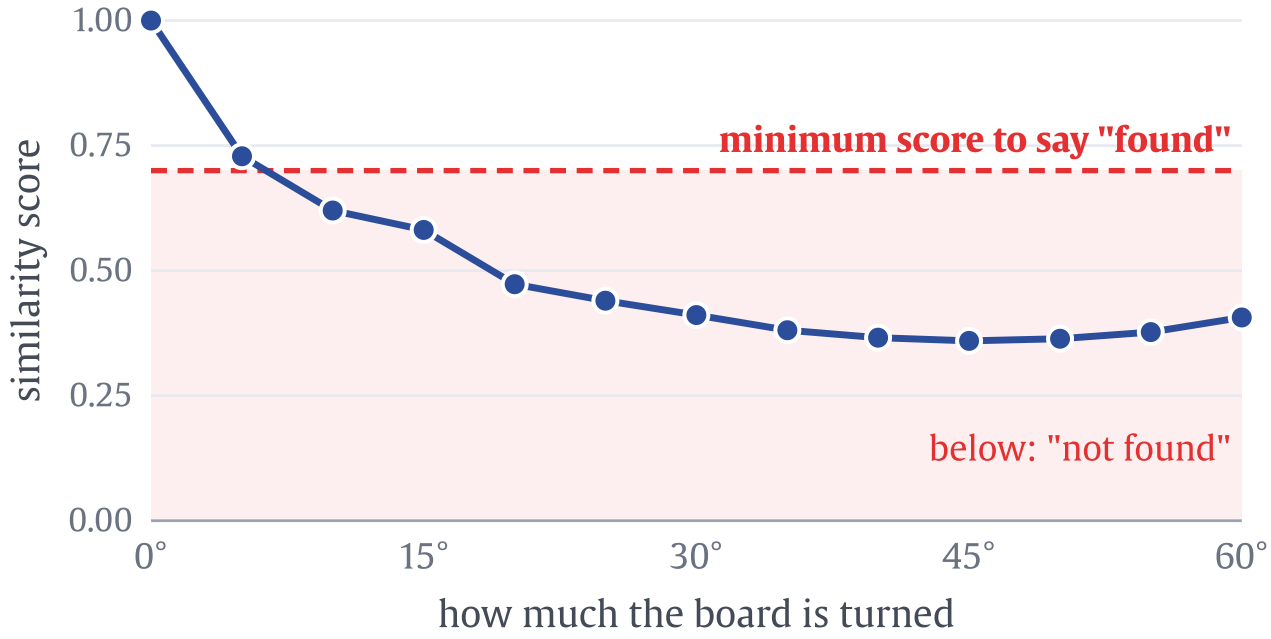


The template (from photo 1)



The same chip, board turned by 30°

It is the same chip — but pixel by pixel the two pictures no longer match: the score falls from 1.00 to **0.41**.

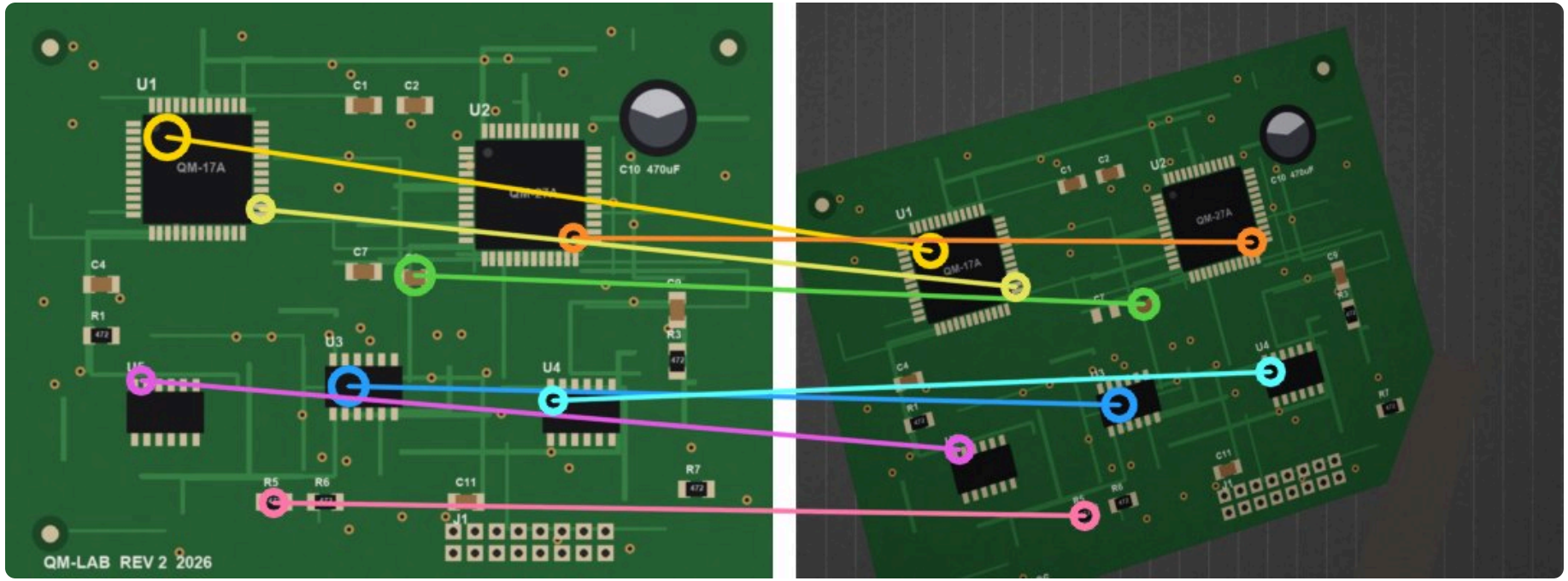


The software says “found” only above a minimum score. Turned by just 10°, the score is already 0.62: the chip is reported **missing although it is there**.

We need something that does not change
when the board turns, moves farther away or gets darker.

Idea 2: find interest points

A few distinctive spots, found again in both photos



Distinctive
It can be told apart from the rest of the board.

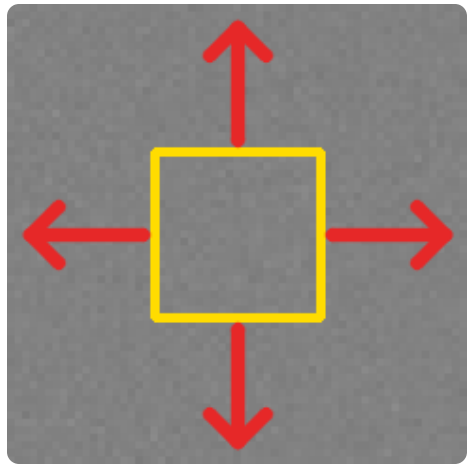
Found again
Whatever the angle, the size and the light.

Precise
It has one exact position.

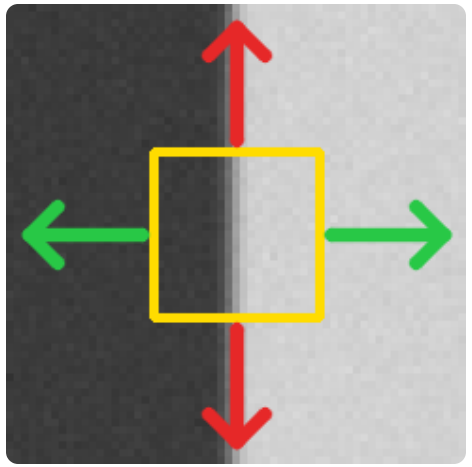
If we know where a few interest points went, we know where the **whole board** went. Like landmarks on a map, they tell you where you are.

What makes a good interest point?

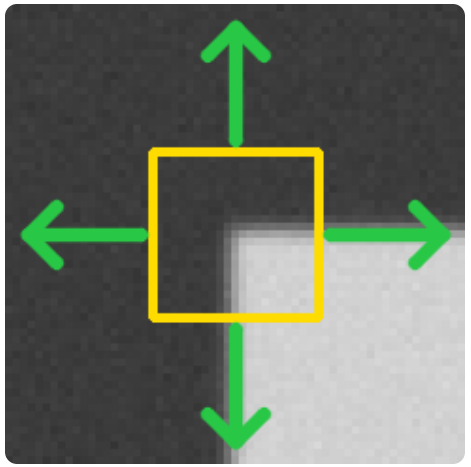
Move a small window a little: does what you see change?



Plain area — nothing changes, whichever way you move. **Cannot be found again.**



Edge — changes across it, not along it. **The position slides along the edge.**



Corner or spot — changes whichever way you move. **One exact position: a good interest point.**

moving this way changes the view moving this way changes nothing

EVERYDAY

A white wall, a long fence, the corner of a window frame: which one could you point to again, exactly?

So far TAKEAWAYS

Part 1

1. To a computer a photo is a grid of numbers — and when the board turns, moves away or gets darker, all of them change.

2. Comparing whole pictures (template matching) works only if the part looks exactly the same as the stored picture.

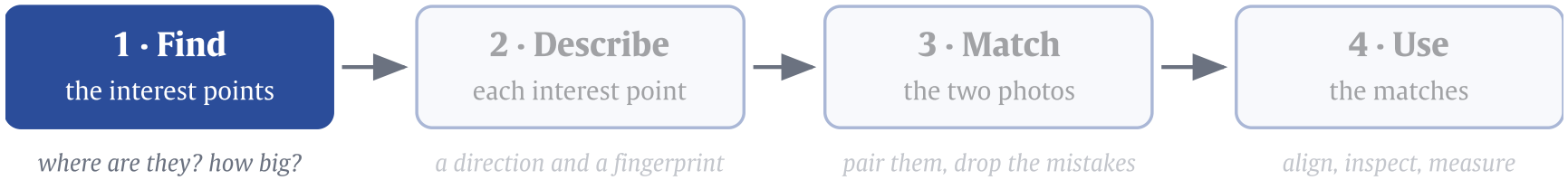
3. Instead: find a few **interest points** in both photos. Good interest points are corners and spots — not plain areas, not edges.

Next question
How does a computer find such spots — and at what size, if the board can be near or far?

PART 2 · STEP 1

Find the interest points

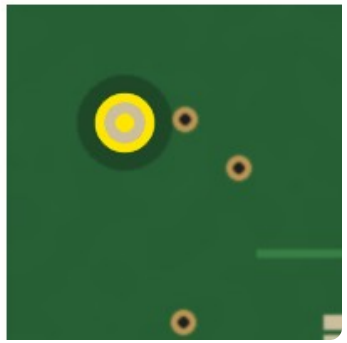
Where are they, and how big?



Spots, or “blobs”

The kind of interest point the method looks for

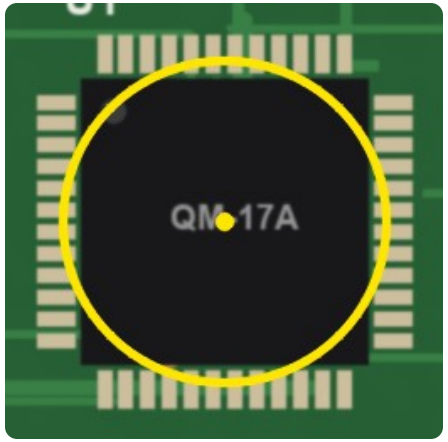
A **blob** is a spot that is brighter or darker than what is around it: a solder pad, a capacitor, a hole, a screw head. It is described by two things: **where its centre is** and **how big it is**.



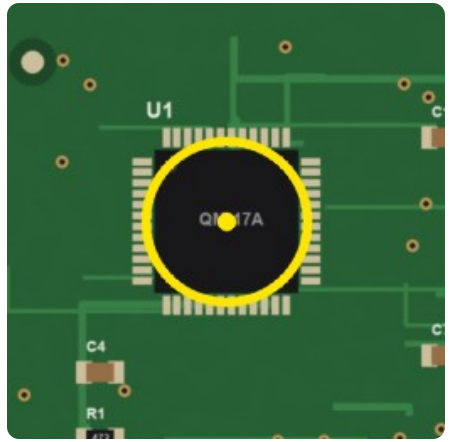
fiducial dot



capacitor



chip

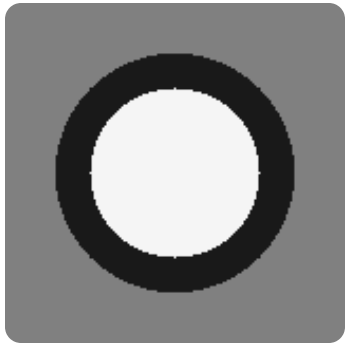


the same chip, from twice as far

> **The catch**
> How big a blob looks depends on how far the camera is. So the computer must find blobs of **every** size — and measure the size of each.

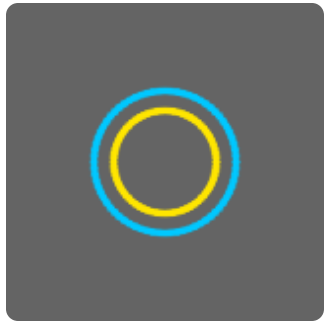
A stencil for spots: middle minus ring

A score you could compute by hand

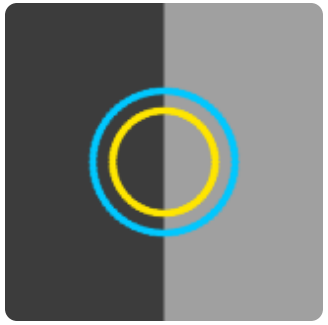


White: the **middle**. Black: the **ring** around it. On the examples: inside the yellow circle, and between yellow and cyan.

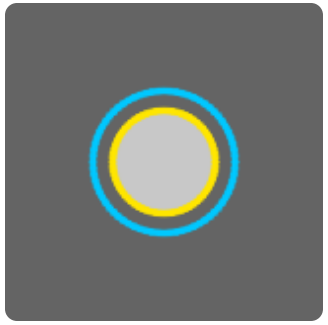
Score = average brightness of the middle − average brightness of the ring



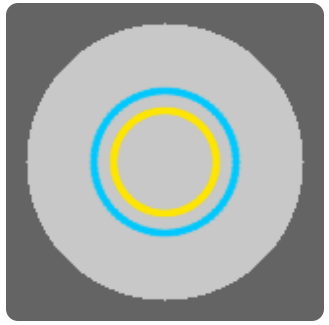
Plain area
100 − 100 = **0**



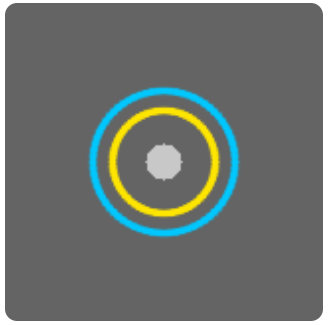
An edge
110 − 110 = **0**



A spot of the right size
200 − 100 = **+100**



A spot much bigger
200 − 200 = **0**

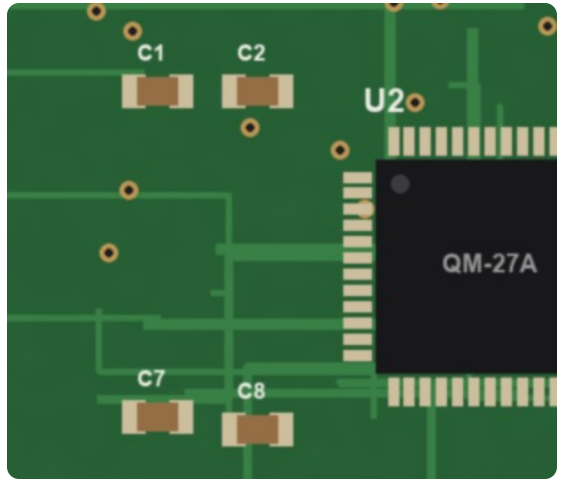


A spot much smaller
112 − 100 = **+12**

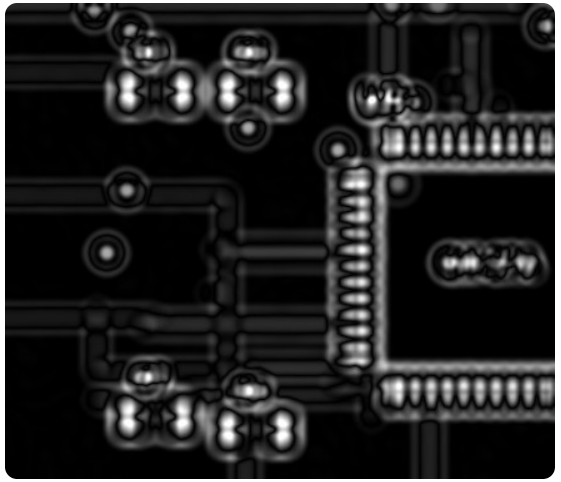
Only a spot of the right size scores high
Plain areas and edges give zero — the bad interest points of Part 1 are ignored automatically.

Slide the stencil over the whole board

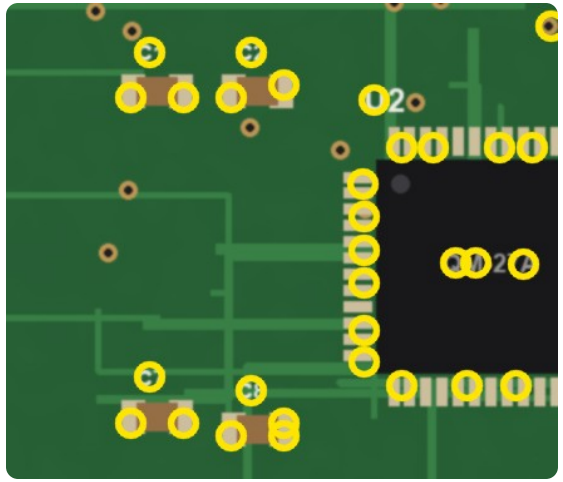
At every position, compute middle minus ring



1 · part of the board



2 · the score at every position (bright = high)



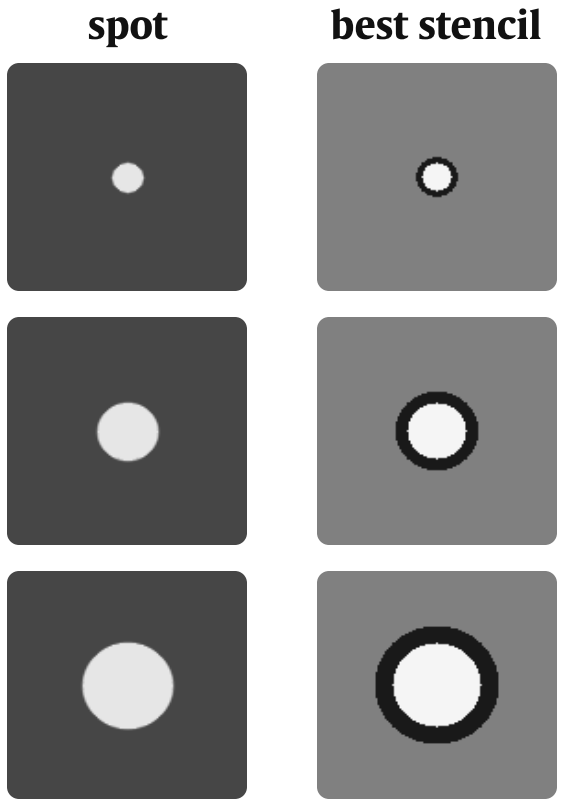
3 · the highest scores: spots found

A dark spot on a bright background scores just as strongly, with a minus sign: it counts too.

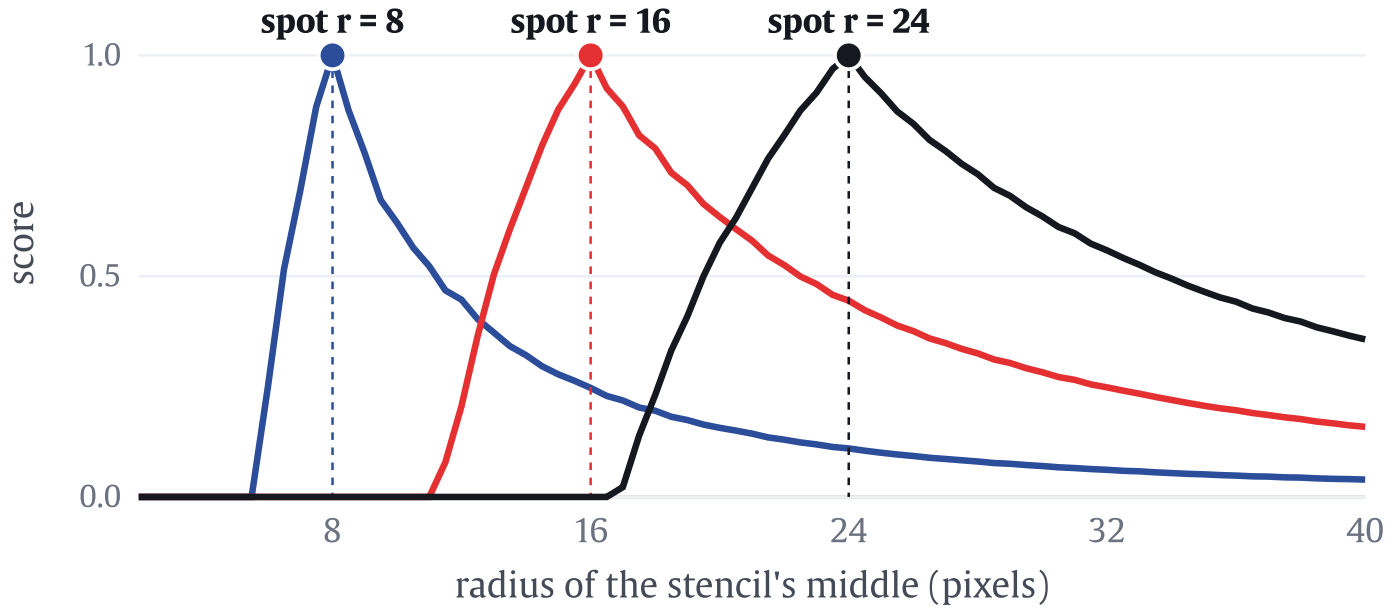
> **Like the template of Idea 1 — with one difference**
> The stencil is **round**, so it is the same at any angle: turning the board changes nothing.

The stencil that scores best tells the size

Try stencils from small to large on each spot



Same scale. The best stencil is the one whose **middle is exactly as big as the spot**.



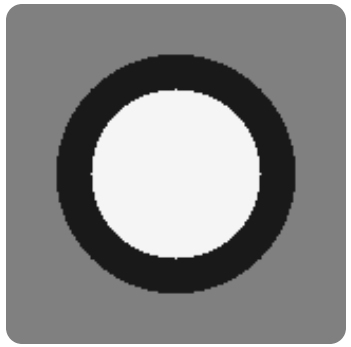
- Stencil too small: its ring is still inside the spot → middle − ring = 0.
- Stencil too large: the spot fills only part of the middle → low score.
- Just right: middle all spot, ring all background → the highest score.

Size is measured, not assumed

Try every stencil size: the one that scores best gives the size of the spot. That is what lets the method find the board near or far.

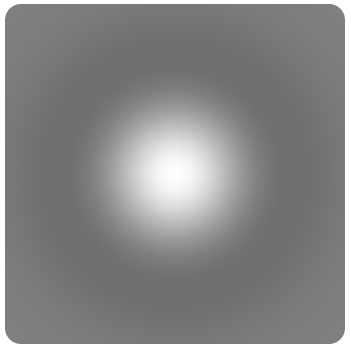
In practice: a smooth stencil

Its name is “Laplacian of Gaussian”



sharp: middle and ring

→



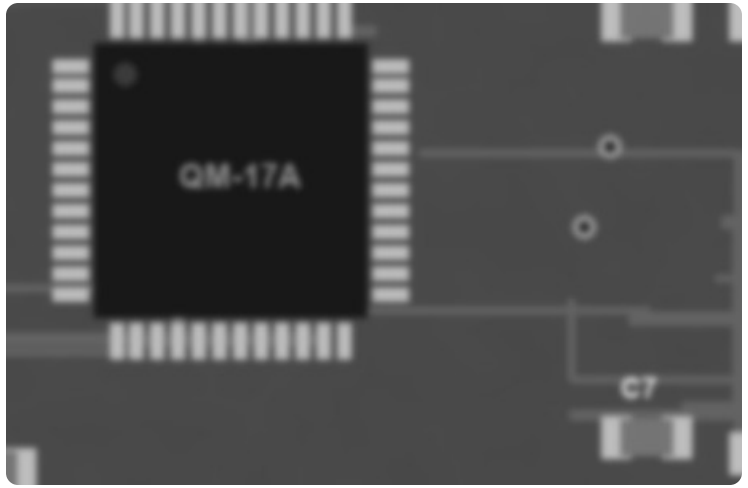
smooth: the same, with soft edges

- A sharp edge between middle and ring reacts to every bit of noise. The smooth version does the same job, with **steadier scores**.
- Its size is called **σ** (sigma): the middle has a radius of about 1.4 σ.
- **“Gaussian”**: it is built from the bell curve of Lecture 6. **“Laplacian”**: it measures how much a point stands out from its surroundings — middle minus ring.

On our board, the fiducial dot (radius 13 pixels) scores best at σ ≈ 9: a middle of radius 1.4 × 9 ≈ 13 pixels — exactly its size.

All sizes at once: blur more and more

“Scale space” — like stepping back from the board, one step at a time



A little blur: the pins are still there.



More blur: pins gone, parts remain.



A lot of blur: only the chip remains.

Blurring with a wider bell is like looking from farther away: **small details disappear first**, big ones last. The computer makes a whole ladder of blurred copies of the photo — one for each stencil size.

Subtract two neighbouring blur levels

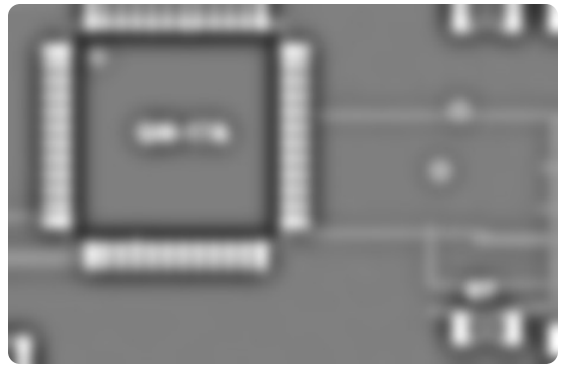
What the extra blur wipes out are the spots of one size



blurred

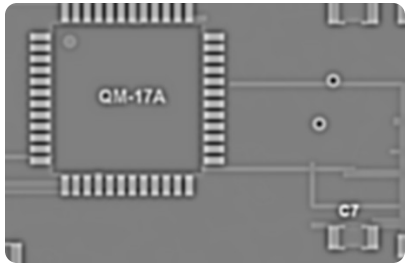


blurred a bit more

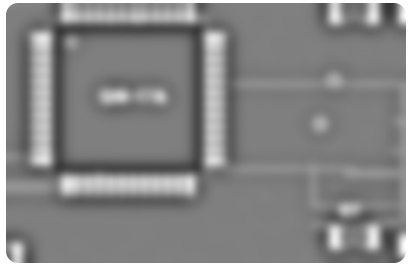


difference: details of one size stand out

The difference keeps only what the extra blur wiped out: each spot minus its blurred surroundings — middle minus ring, the smooth stencil again, for free. Each pair of levels gives the spots of one size:



small spots: pins, vias



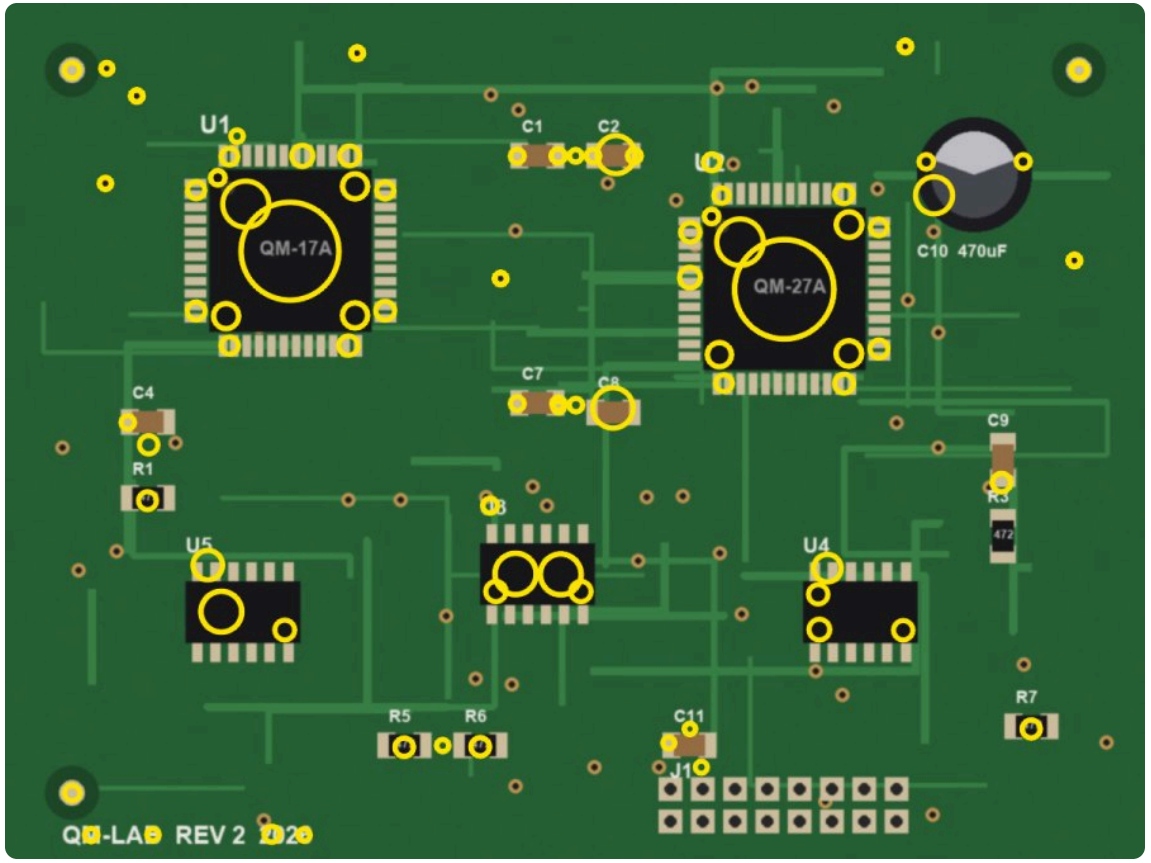
medium spots: pin rows, capacitors



large spots: the chip

Step 1 done: the interest points of our board

SIFT — Scale-Invariant Feature Transform: "scale-invariant" means it does not care about size



Some of the interest points found (the larger ones). Circle = centre and size.

On the reference board	spots left
Spots found, at every size	7,289
– too faint: probably noise	3,315
– lying on an edge	2,412

2,412 interest points, each with a centre and a size. Photo 2 gives 1,979.

> **Next question**
> Thousands of interest points in each photo. Which one in photo 1 is which one in photo 2?

Recap TAKEAWAYS

Step 1 — find the interest points

1. The method looks for **blobs**: spots brighter or darker than their surroundings, each with a centre and a size.
2. A **round stencil** finds them: score = middle minus ring. Plain areas and edges score zero; being round, the angle does not matter.
3. The stencil size that scores best **measures** the size of the blob — so the board can be near or far.
4. **Blurring step by step** searches all sizes at once; faint spots and edges are thrown away.

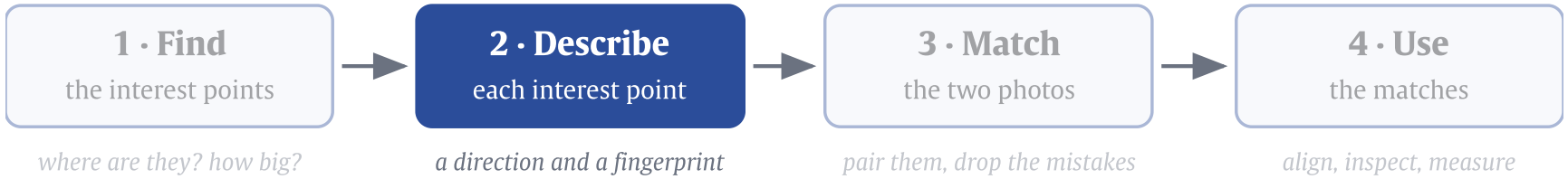
DISCUSS

A tray of identical washers: every washer is a perfect blob. Why will that be a problem when we try to tell which washer is which?

PART 3 · STEP 2

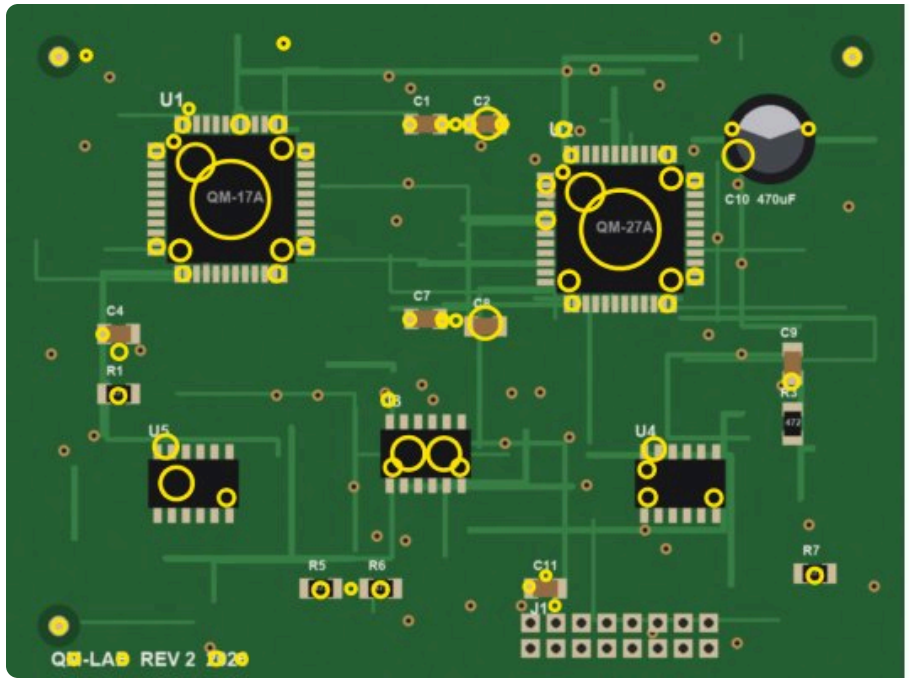
Describe each interest point

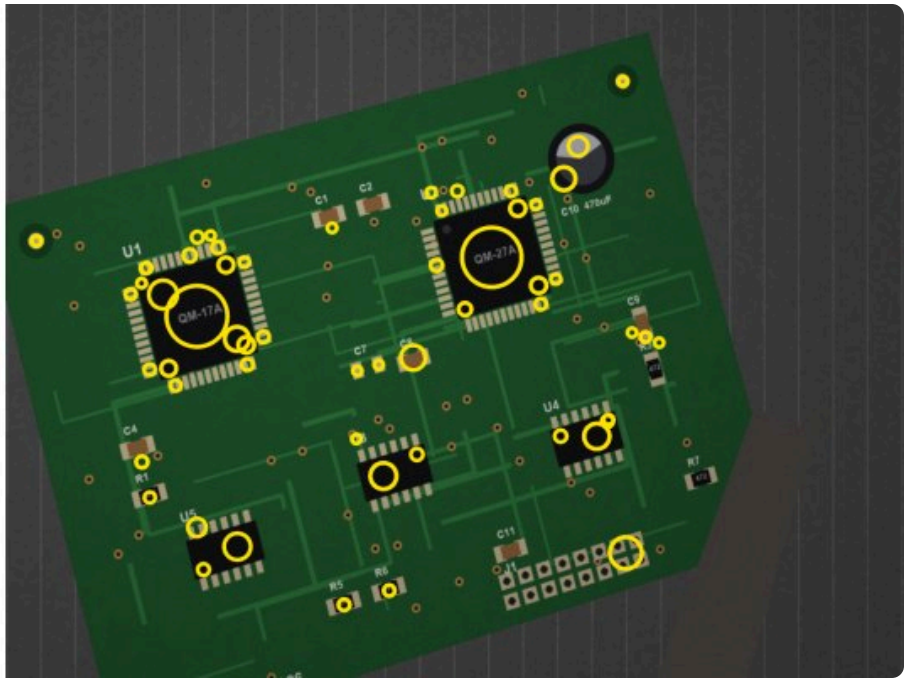
A description that stays the same when the board turns, moves away or gets darker



Which interest point is which?

The interest points of both photos — but nothing yet says which belong together





Each interest point needs a **fingerprint**: a description of what it looks like. Two requirements:

The same spot → the same fingerprint
even if the board is turned, smaller or darker in the other photo.

Different spots → different fingerprints
so that no interest point is mistaken for another.

First, a direction for every interest point

Turn each one upright and to the same size, and the difference between the photos disappears

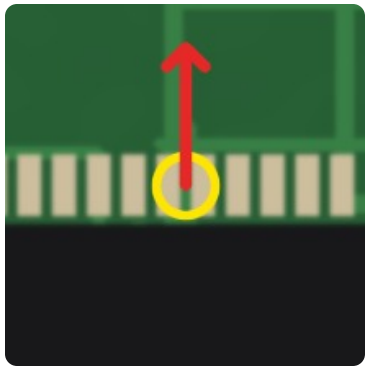


Photo 1, as seen

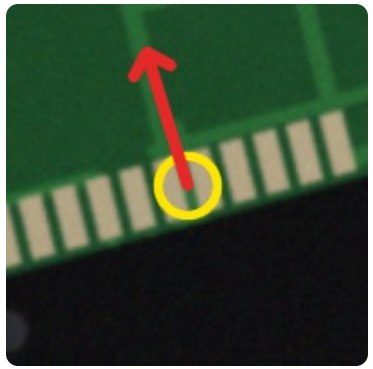


Photo 2, as seen: turned and smaller

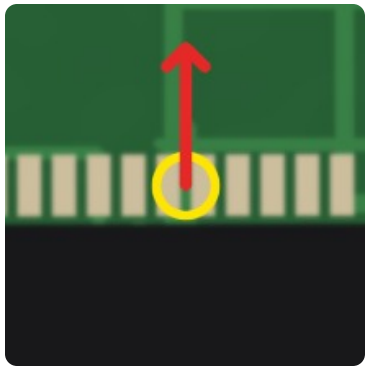


Photo 1, turned upright

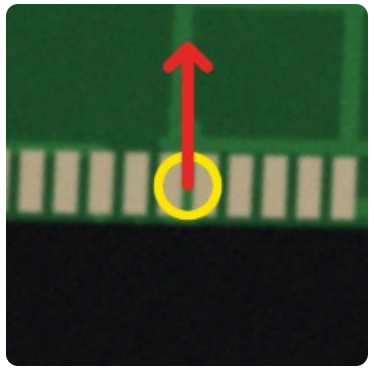


Photo 2, turned upright

Around each interest point, find the direction in which the brightness changes most: its **compass needle** (red arrow). Turn the patch so the needle points up, and scale it to the interest point's size (yellow circle). The same spot from the two photos now **looks the same**, and it is written down as 128 numbers: its **fingerprint**.

What the fingerprint records

The upright patch is cut into 4 × 4 small squares. For each square: how strongly the brightness changes in each of 8 directions. 16 × 8 = 128 numbers — in words, **where the edges are around the point and which way they run**.

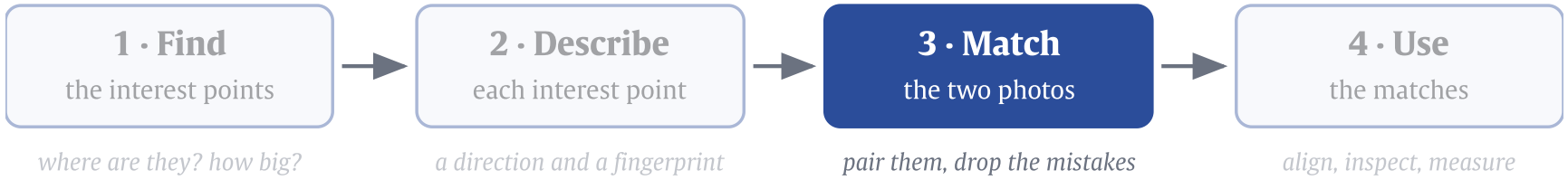
What it leaves out, on purpose

The angle and the size (already removed by turning and scaling), the overall brightness (the numbers are scaled to a fixed total) and the exact pixel values. So the same spot gives the same fingerprint in both photos.

PART 4 · STEP 3

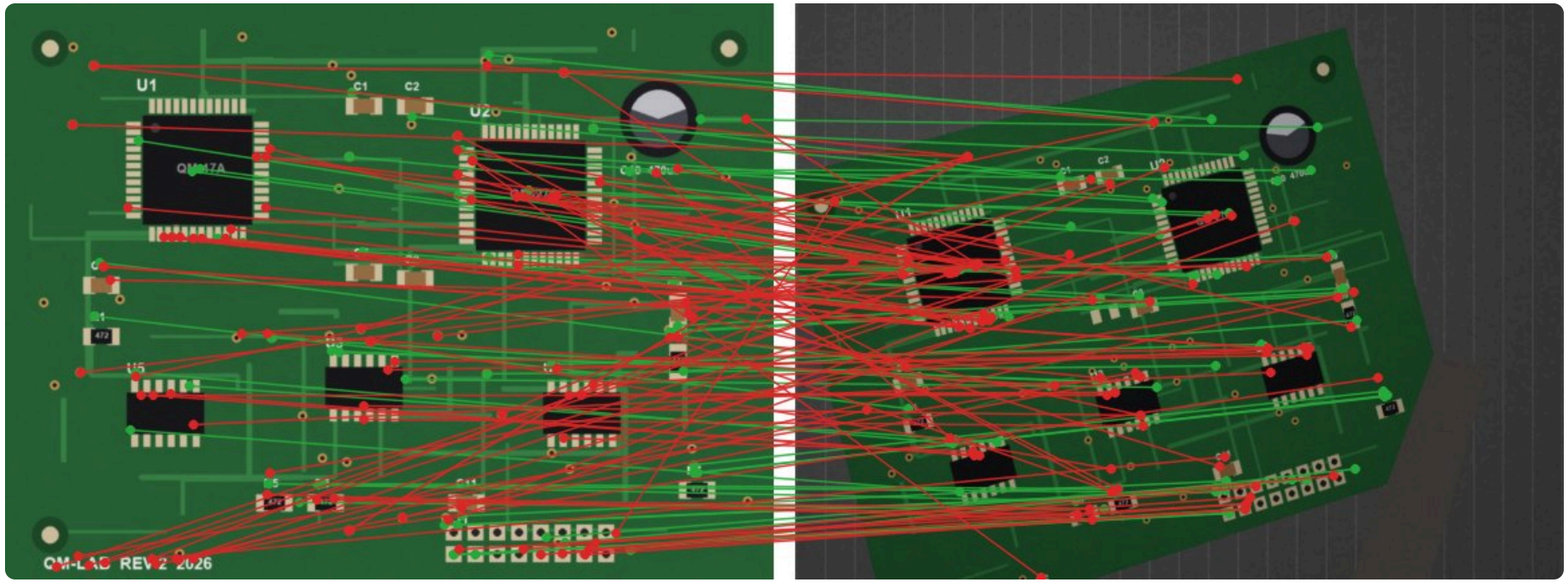
Match the two photos

Pair the interest points, and throw out the mistakes



Pair the most similar fingerprints

Simple — but many pairs are wrong

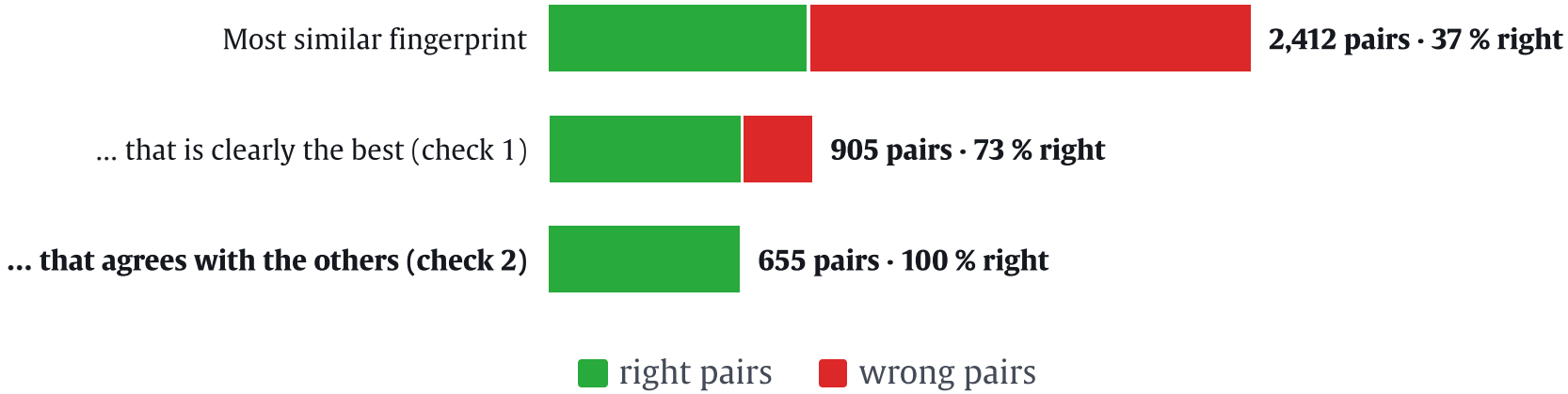


right pair wrong pair (150 of 2,412 drawn)

For every interest point of photo 1, take the interest point of photo 2 with the closest fingerprint. Only **37 %** of the pairs are right. Why? Many interest points look alike — 48 identical pins on each chip, identical capacitors — and some are not visible at all in photo 2.

Two checks remove the wrong pairs

From 2,412 pairs to 655 — and all of them right



Check 1 · Is it clearly the best?

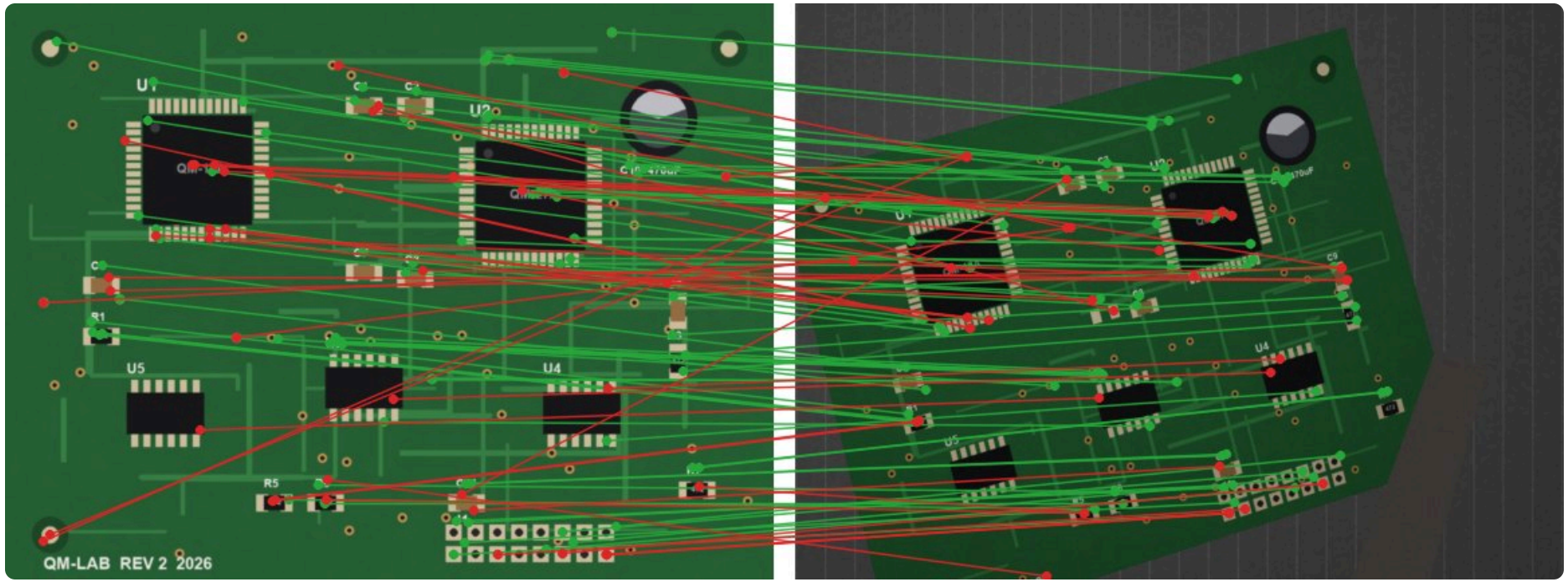
If a second candidate is almost as similar — one of the 48 identical pins of a chip — the pair is a guess. Drop it.

Check 2 · Does it agree with the others?

The board moves as one piece, so every right pair shows the **same** turn, shift and size change. Pairs that tell a different story are dropped.

The board, found

Turned, smaller, darker, partly hidden — and located to a fraction of a pixel



pairs that agree (70 of 655 drawn) pairs dropped by check 2 (40 of 250 drawn)

From the 655 agreeing pairs, the computer works out how the board moved. The result is within **0.17 pixels** of the true movement, everywhere on the board — better than template matching ever did, and with no stored picture at all.

Recap TAKEAWAYS

Steps 2 and 3 — describe and match

1. Each interest point gets a **direction**; turning it upright and to its size removes the rotation and the distance.
2. Its **fingerprint** (128 numbers) describes which way the edges run around it, independently of the light.
3. Pairing the closest fingerprints gives many wrong pairs, because repeated parts look alike.
4. Two checks remove them: keep clear winners only, and only pairs that agree on **one movement** of the board.

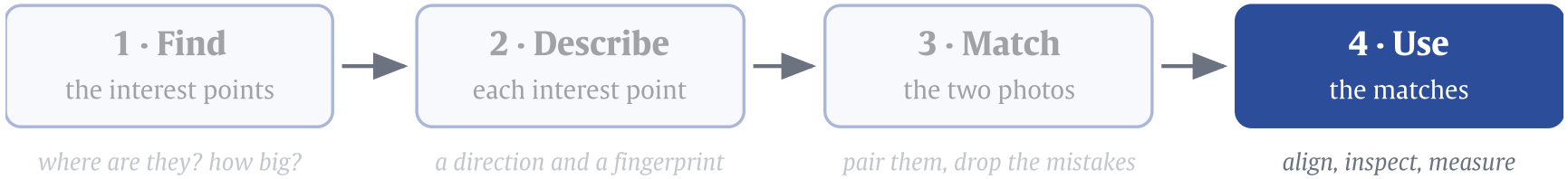
DISCUSS

Only 40 pairs pass check 1, and check 2 rejects 35 of them. What would you suspect — the board, the light, or the camera?

PART 5 · STEP 4

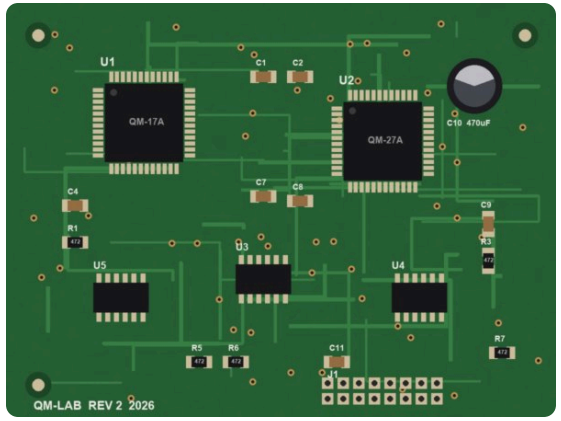
Use the matches

Back to the question: is anything missing on the board?

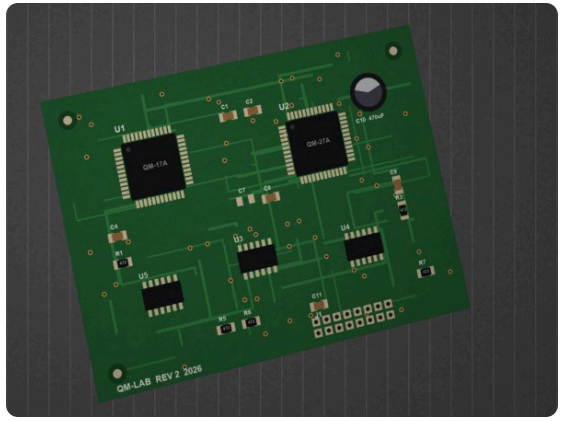


Align, then compare

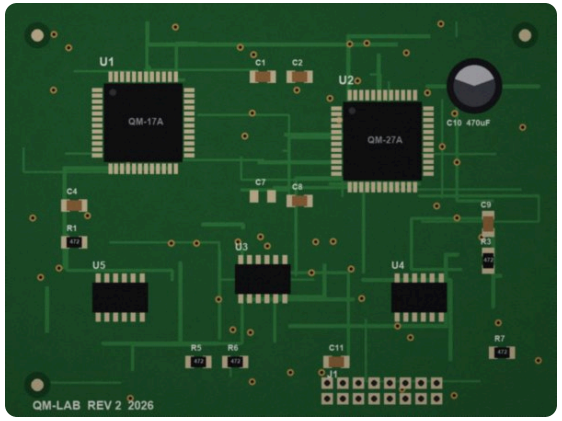
The answer to today's question



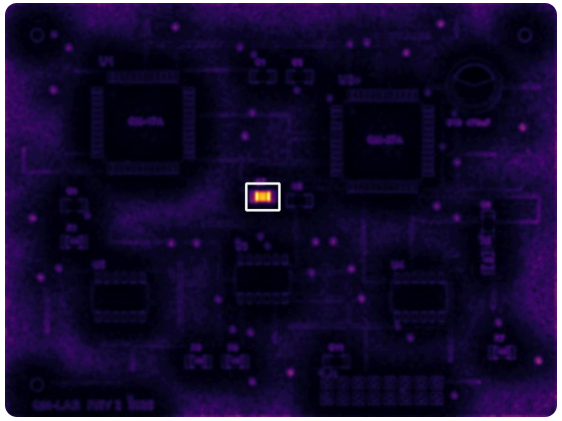
1 · the reference



2 · a board on the line



3 · turned back, using the matches



4 · the difference

The matches (631 agreeing pairs) tell how the board moved. Turning photo 2 back puts every part exactly on top of the reference. Subtract the two, and only one spot remains: **capacitor C7 is missing**.

This is how automated optical inspection (AOI) of circuit boards works. The fiducial dots in the corners are interest points put there on purpose, so they can always be found.

Stitching: one picture from several photos

When the part is too large for one photo

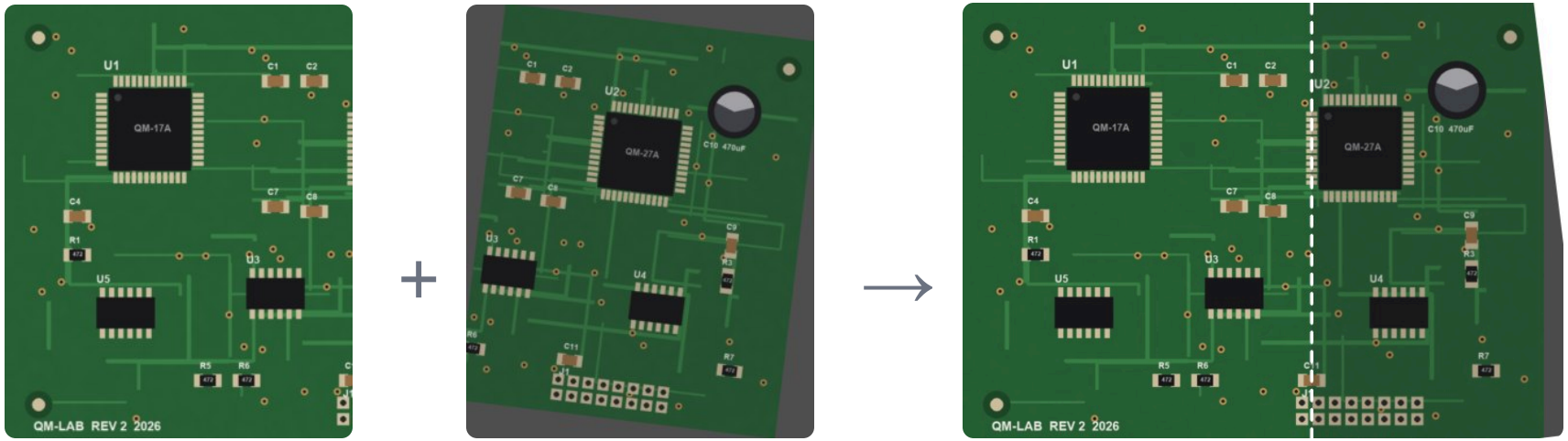


Photo 1: the left part

Photo 2: the right part, turned, darker

Stitched. Dashed line: where photo 1 ends.

Interest points in the overlap (149 agreeing pairs) tell how photo 2 sits relative to photo 1. Photo 2 is turned to fit, and the two are joined.

EVERYDAY · THE PANORAMA ON YOUR PHONE

As you sweep the phone, it does exactly this between consecutive frames. Visible seams are wrong or missing matches.

INDUSTRY

Car doors, aircraft panels, wind-turbine blades seen by a drone: a camera close enough to see small defects sees only part of them.

Where else interest points do the work

The same four steps, different last step

Guiding a robot
Where is the part, and which way is it turned? The robot picks it up the right way round — parts in a bin, bodies on a car line.

Recognising and sorting
Which part, which variant, which label is in front of the camera? Mixed parts on one conveyor are sent to the right station.

Measuring in 3D — next lecture
The same interest point seen by two cameras from two places: the difference between the two views gives its distance. That is stereo vision.

Following motion
The same interest points in every frame of a video show how a part moves or vibrates — the subject of this course's internship project.

When there are no interest points

The method needs something to find

Where it struggles

- **Plain or shiny surfaces**: nothing to find.
- **Repeated patterns**: everything looks alike.
- **Motion blur**: the pattern is smeared.
- **Bending parts**: no single movement fits all pairs.

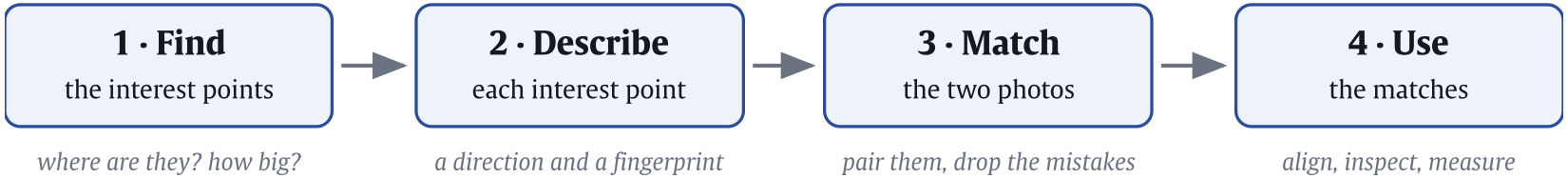
Then: put interest points there

- **Fiducial dots** on circuit boards.
- **Coded targets** stuck on large parts for 3D measurement.
- A random **speckle pattern** sprayed on a part, to follow how it deforms.

SIFT is one recipe among several; faster ones (for example ORB) run on phones and small robots. The four steps are always the same.

The whole story TAKEAWAYS

One board, two photos, four steps



1. **Find:** spots that stand out, with a round stencil at every size — so angle and distance do not matter.
2. **Describe:** turn each spot upright, to its size, and record which way its edges run: 128 numbers.
3. **Match:** pair similar fingerprints, keep clear winners that agree on one movement.
4. **Use:** align and compare, stitch, guide a robot, measure in 3D.

Next lecture — 3D optical shape measurement: the same interest point seen by two cameras gives its distance.

Questions?

Prof. Gianluca Rossi • dr. Lorenzo Capponi

lorenzo.capponi@collaboratori.unipg.it • lorenzocapponi@outlook.it

github.com/LolloCappo/student-hub